Name: ______________________

Worksheet

Solution to an equation

Task A: Testing solutions to equations

1) Fill in the table for both expressions. Use your table to find the **solution** to the equation.

a) equation

$2x + 3 = 4x - 5$

	$x = 1$	$x = 2$	$x = 3$	$x = 4$	$x = 5$
$2x + 3$					
$4x - 5$					

solution

b) equation

$3(x - 4) = 5x$

	$x = 2$	$x = 0$	$x = -2$	$x = -4$	$x = -6$
$3(x - 4)$					
$5x$					

solution

c) equation

$20 - 2x = 2x + 10$

	$x = 1$	$x = \frac{3}{2}$	$x = 2$	$x = \frac{5}{2}$	$x = 3$
$20 - 2x$					
$2x + 10$					

solution

2) Sort these equations into the correct column in the table.

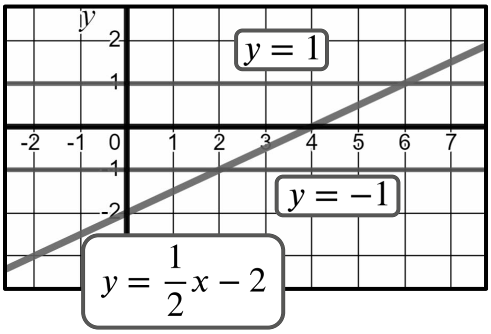
has $x = 2$ as a **solution**	has $x = 0$ as a **solution**	has no **solution**

$5x = 10$ $\quad x = 3x$ $\quad 5(x + 1) = 15$ $\quad 2 - x = 7 - x$ $\quad 3(2x + 5) = 6x - 2$

$3x - 6 = 0$ $\quad 2x + 5 = 2x + 3$ $\quad 2x - 2 = 6 - 2x$ $\quad x - 2 = 10x - 2$

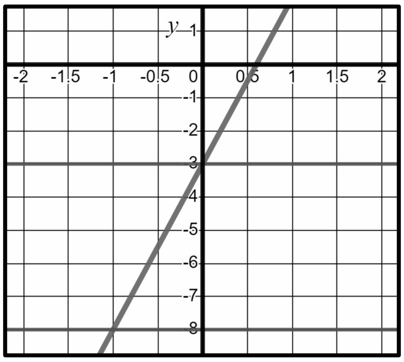
Task B: Finding solutions from graphs

1a) Use the lines on the graph to solve the equations below.



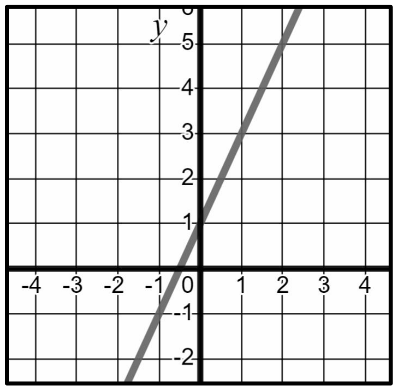
equation	**solution**
$\frac{1}{2}x - 2 = 1$	
$\frac{1}{2}x - 2 = -1$	

1b) Below is the graph of the equation $y = 5x - 3$ Use it to solve the equations.



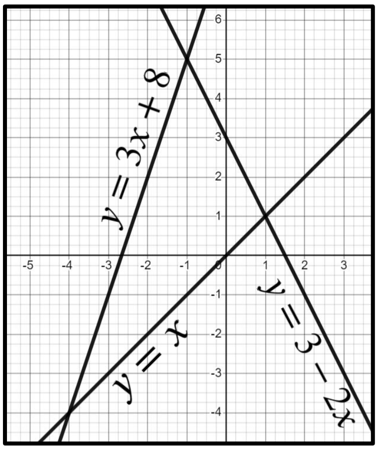
equation	**solution**
$5x - 3 = -3$	
$5x - 3 = -8$	

1c) Solve the equations below by drawing the relevant graph.



equation	**solution**
$2x + 1 = 3$	
$2x + 1 = 1$	

2) Use the lines on the graph to solve the equations.



a)

equation	**solution**
$x = 3 - 2x$	

b)

equation	**solution**
$x = 3x + 8$	

c)

equation	**solution**
$3x + 8 = 3 - 2x$	

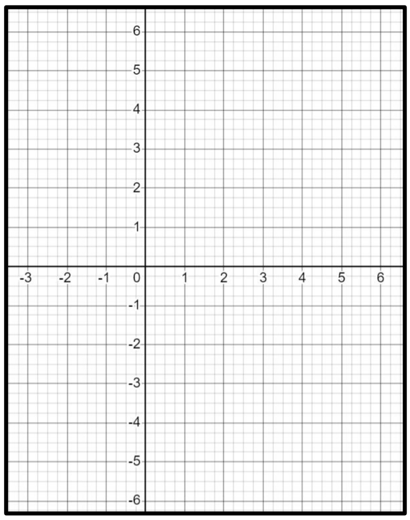
3) Solve the equation below by drawing the relevant graphs.

$y = 3x - 3$

x	−1	0	1	2	3
y					

$y = 5 - x$

x	−1	0	1	2	3
y					



equation	**solution**
$3x - 3 = 5 - x$	